

Scattering

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Data

- (a) $L \in \mathbb{N}$ (dimension).
- (b) Numbers $a_1 \geq a_2 \geq \dots \geq a_L > 0$.
- (c) $N, M \in \mathbb{Z}$ with $N < M$ (minimum and maximum of frame).
- (d) Finitely many self-adjoint potentials $V(j_1), \dots, V(j_K) \in \text{Mat}(L \times L, \mathbb{C})$, $j_1, \dots, j_K \in \mathbb{Z}$, $N \leq j_1 < \dots < j_K \leq M$.
- (e) An interval $[a, b] \subseteq [-2a_L, 2a_L]$.
- (f) Functions $f_{l,\sigma} : [a, b] \rightarrow \mathbb{C}$, $l = 1, \dots, L$, $\sigma = \pm$.

Algorithm

- (1) For $l = 1, \dots, L$ and $E \in [a, b]$, solve the equation $z^2 - \left(\frac{E}{a_l}\right)z + 1 = 0$. There will be two complex-valued solutions. We denote the solution with $\text{Im}(z) > 0$ by $z_l(E)$.
- (2) For $n \in \mathbb{Z} \cap [N, M]$, compute

$$\phi^\pm(n, E) := \text{diag}(z_l(E)^{\mp n})_{l=1, \dots, L} \in \text{Mat}(L \times L, \mathbb{C}).$$

- (3) For $l = 1, \dots, L$ and $E \in [a, b]$, compute

$$\nu_l(E) := \frac{1}{2a_l \text{Im}(z_l(E))}.$$

(4) For $n \in \mathbb{Z} \cap [N, M]$, $k = 1, \dots, K$, and $E \in [a, b]$, compute

$$s(n, k, E) := \text{diag}(\nu_l(E) [z_l(E)^{j_k - n} - z_l(E)^{n - j_k}])_{l=1, \dots, L} \in \text{Mat}(L \times L, \mathbb{C}).$$

(5) For $n \in \mathbb{N} \cap [N, M]$ and $E \in [a, b]$, compute inductively

$$u_+^{E, \pm}(n) = \phi^\pm(n, E) \pm \sum_{\substack{k=1, \dots, K \\ j_k > n}} s(n, k, E) V(j_k) u_+^{E, \pm}(j_k) \in \text{Mat}(L \times L, \mathbb{C})$$

and

$$u_-^{E, \pm}(n) = \phi^\mp(n, E) \mp \sum_{\substack{k=1, \dots, K \\ j_k < n}} s(n, k, E) V(j_k) u_-^{E, \pm}(j_k) \in \text{Mat}(L \times L, \mathbb{C}).$$

(6) For $\sigma, \tau \in \{\pm\}$ and $E \in [a, b]$, compute

$$W_\tau^{E, \sigma} := \left[u_{-\tau}^{E, -\sigma}(N+1)^* A u_\tau^{E, \sigma}(N) - u_{-\tau}^{E, -\sigma}(N)^* A u_\tau^{E, \sigma}(N+1) \right] \in \text{Mat}(L \times L, \mathbb{C}).$$

(7) For $n \in \mathbb{Z} \cap [N, M]$, $k = 1, \dots, K$, $\sigma \in \{\pm\}$, and $E \in [a, b]$, compute the matrices

$$G^{E, \sigma}(n, k) = \begin{cases} u_+^{E, \sigma}(n) \left(W_+^{E, \sigma} \right)^{-1} u_-^{E, -\sigma}(j_k)^* & \text{if } n > j_k \\ -u_-^{E, \sigma}(n) \left(W_-^{E, \sigma} \right)^{-1} u_+^{E, -\sigma}(j_k)^* & \text{if } n \leq j_k. \end{cases}$$

(8) For $n \in \mathbb{Z} \cap [N, M]$, $l = 1, \dots, L$, $\sigma \in \{\pm\}$, and $E \in [a, b]$, compute

$$w_{l, \sigma}^{E, \pm}(n) := z_l(E)^{\mp n} e_l - \sum_{k=1}^K G^{E, \sigma}(n, k) V(j_k) z_l(E)^{\mp j_k} e_l \in \mathbb{C}^L.$$

(9) Compute

$$p = \sum_{\substack{l=1, \dots, L \\ \sigma=\pm}} \int_a^b |f_{l, \sigma}(E)|^2 dE.$$

(10) For $n \in \mathbb{Z} \cap [N, M]$ and $t \in \mathbb{R}$, compute

$$\psi(n, t) := \frac{1}{p} \left| \sum_{\substack{l=1, \dots, L \\ \sigma=\pm}} \int_a^b e^{-itE} f_{l, \sigma}(E) w_{l, \pm}^{E, \sigma}(n) \sqrt{\nu_l(E)} dE \right|^2.$$